On the study of fundamental modes of spherical dielectric resonator at millimeter-wave frequency

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***Abstract*—The information-rich canonical problem of electromagnetic wave interaction with dielectric spheres has been explored for over a century. This has revolutionized several state-of-art technologies in the optical frequency regime. However, such principles have not been extensively scaled down to frequencies of operation in electrical circuits. With the futuristic demand for consumer electronics subservient to millimeter-waves, the large-scale use of dielectric-based circuit elements is self-evident. As far as mass production of dielectric bodies is concerned, a spherical geometry will provide the most accessible fabrication with minimum tolerance and cost compared to other generic shapes of dielectric bodies. Hence, in this context, the current paper explores electromagnetics involving homogeneous dielectric spheres from the perspective of its potential application to millimeter-wave circuit design. Its fundamental resonance modes are analyzed in the light of Rayleigh scattering and Mie theory, followed by its testing and evaluation in the Ka-band (26.5 GHz to 40 GHz). Finally, the concept of an on-chip resonator based on the spherical magnetic dipole mode is demonstrated.**

***Index Terms*—dielectric resonator, electric dipole, electromagnetic scattering, magnetic dipole, Mie theory, on-chip resonator.**

I. Introduction

The interaction of electromagnetic waves with matter, encompassing both elastic and inelastic scattering, especially at optical frequency, has always been of significant interest to the scientific research community. The principle of inelastic scattering, like the Raman effect, has been exhaustively applied for material spectroscopy. On the other hand, electromagnetic elastic scattering from multi-wavelength-sized scatterers features some unique phenomena, e.g., whispering gallery modes and photonic nanojets, which has led to the evolution of sensitive sensor design and high-resolution imaging techniques. Alternatively, electromagnetic elastic wave-matter interaction in sub-wavelength-sized bodies is of interest for compact circuit engineering. Potential applications demonstrated at optical frequencies include directive nanoantennas (see e.g., [1]), tunable metasurface, etc.

With such a plethora of state-of-art techniques based on electromagnetic scattering demonstrated at optical frequency, one is only left to ponder over their potential scaling to electrical design, especially in the microwave regime. In this context, several bottlenecks are critical, if not obvious. Firstly, all the instances of inelastic scattering from naturally occurring materials appearing from and above infrared frequencies cease to exist in the microwave regime [2]. Secondly, microwave to terahertz circuit design features significant conduction loss from metal and metal-based circuit elements. Unlike optics, a generic electrical circuit design is bounded to limited power; hence, metal scatterer-based applications are not a suitable option. Moreover, the plasmonic effect of metal scatterer that is widely explored at optical frequencies also does not appear at microwave frequency [3]. Thirdly, the wavelength-based geometric scaling of multi-wavelength-sized scatterer used for optical circuit design will lead to an impractically voluminous size of the circuit element at microwave frequency. This will not lead to a compact, robust, and integrated circuit design. Finally, one must also contemplate the prospective materials available with a wide range of optical properties in addition to their exotic tunable features [4]. However, such a dispersive range of electrical properties (complex dielectric permittivity), especially in the microwave regime, is not readily available for circuit manufacturing. Typical examples in this category include fine-tuned nanoparticles used in optical frequency, which cannot be manufactured and conformed to a bulk volume as required at microwave frequency.

Considering all the above limitations, one of the best possible options to investigate for microwave circuit design will be a sub-wavelength-sized dielectric scatterer, subject to its low-cost and wide range (dimensions and dielectric constant) commercial availability. Such use of dielectric bodies for high-frequency electrical circuit operation was first proposed by Richtmyer in 1939 [5]. The use of dielectric bodies for microwave engineering has matured over the past few decades, and its widespread usage is in the form of cylindrical, cubical, hemispherical topology, and some unique cases of hybrid geometries [6], [7]. The working principle of such dielectric resonators is primarily explored in the context of radiation for antenna design and mutual coupling for filter design. This has resulted in their engineering and applications being widely established in the fields of telecommunications (e.g., base station filters and antennas), satellites (e.g., cavity filters), and consumer electronics (e.g., oscillators).

Among such modern-day electronic technologies, one particular application that uniquely employs a spherical-shaped

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dielectric resonator is the case of tunable filters and oscillators where Yttrium Iron Garnet (YIG)-dielectric resonators are applied [8], [9]. This is possibly the unique case of using a spherical-shaped dielectric resonator and is based on the natural electromagnetic modes of a spherical dielectric scatterer. To date, such spherical dielectric resonator-based tunable circuits are indispensable tools for commercial high-frequency test and instrumentation modules. The futuristic commercial applications and market demands are based on millimeter-wave (mmWave). In this context, there is a surge to engineer dielectric resonators and dielectric waveguide-based circuit design to mitigate the effect of conduction loss at such frequencies. This has led to the emergence of innovative high-frequency electrical circuits based on electromagnetic scattering from dielectric bodies. In this respect, the current work is based on the electromagnetic scattering properties of spherical dielectric bodies.

Although the electromagnetic wave interaction with a dielectric sphere is canonical problem from the past century, its extensive use for electrical circuit design is hardly noticeable. However, one must contemplate that by virtue of its spherical geometry, a dielectric sphere presents the smallest footprint on a planar aperture (e.g., printed circuit board, thin-film wafer, back-end-of-line layers of a chip, etc.). In recent works, it has been demonstrated that the assembly and packaging of a sphere can be achieved with extreme precision relatively easily by means of a air-filled via/cavity drilled/etched on the planar aperture [10], [11]. Such an attribute is invaluable at mmWave, where the circuit dimensions are in the order of millimeters and micrometers. It enables the integration of dielectric resonators rather than its *in-situ* fabrication, which can be costly and complex. Furthermore, the manufacturing tolerances associated with spherical geometry is single-fold (diameter), whereas dielectric resonators of cylindrical and cuboidal shape will inevitably feature tolerances along multiple axes. As far as its commercial availability is concerned, modern-day manufacturers and suppliers of chemicals and ceramics (e.g., Goodfellow Cambridge Ltd. based in the UK, SITUS Technicals GmbH based in Switzerland, Cospheric based in the USA, etc.) feature dielectric spheres in a wide range of dimensions (from tens of centimeters to a couple of micrometers) and materials (low-permittivity polymers to high-permittivity ceramics) in their catalog.

Taking into consideration all these perks, the potentials of a spherical dielectric resonator for mmWave circuit design are explored theoretically and experimentally in context to its fundamental resonance modes, viz., the electric dipole and the magnetic dipole mode. These modes will be instrumental to the adaptions of the spherical dielectric resonators; hence, this topic is the focus of the current work.

II. Electromagnetic scattering and natural modes of dielectric sphere

On the topic of electromagnetic scattering by spherical scatterers; prior to the publication of the iconic Mie theory in 1908 [12], and the classical works of Debye in 1909 [13], Lord Rayleigh in 1871 published the famous Rayleigh scattering pertaining to the elastic scattering of electromagnetic waves by arbitrarily small particles [14]. The Rayleigh hypothesis is based on electric dipole mode (Fig. 1) induced in dielectric spherical particles operating in the sub-wavelength regime. The mathematical definition of a 'sub-wavelength' sized sphere dictates that the size parameter ($x = 2\pi r/\lambda$, where r is the sphere's radius and λ is the incident free-space wavelength) is less than unity. In this approximation, the Rayleigh scattering efficiency (ratio of scattering cross-section to physical cross-section of the scattering sphere) is given as [15]

$$Q_{sca}^{(Ra)} = \frac{8}{3}x^4\left|\frac{m^2-1}{m^2+2}\right|^2, \tag{1}$$

where m is the ratio of the refractive index of the sphere to the surrounding medium. In the case of a lossless ($\tan\delta = 0$) dielectric sphere of dielectric constant ε_r suspended in free-space, $m = \sqrt{\varepsilon_r}$. As shown in Fig. 2, the Rayleigh scattering efficiency increases linearly with the fourth power of the size parameter, indicating the growing strength of the radiating electric dipole mode with the size of the particle and the operating frequency. Alternatively, Mie theory introduces the concept of an infinite number of electric and magnetic modes, which characterizes the scattering spectra of spherical particles (see Fig. 1). Accordingly, the scattering efficiency is formulated as [15]

$$Q_{sca}^{(Mie)} = \frac{2}{x^2}\sum_{n=1}^{\infty}(2n+1)(|a_n|^2+|b_n|^2), \tag{2}$$

where, a_n and b_n are Mie coefficients for electric and magnetic modes, respectively, of order n. In classical literature and seminal textbooks, the terms electric and magnetic modes are invariably used as transverse magnetic modes (TM) and transverse electric modes (TE), respectively. The fundamental electromagnetic modes are the dipole modes ($n = 1$) and the higher order modes are quadrupole ($n = 2$), octopole ($n = 3$), and so on. Here, it should be noted that the transverse field component of the modes is directed along the radial axis (r) with the co-ordinate system placed at the geometric center of the spherical volume of the dielectric body as shown in Fig. 1. They are exhaustively expressed as E_{nqr} or TM_{nqr}, and H_{nqr} or TE_{nqr}, where the mode number (n, q, r) corresponds to the spherical coordinate system (θ, ϕ, r), respectively. Along the azimuth axis (ϕ), the field variations are irrelevant to any electromagnetic property (resonance, Q-factor, etc.), i.e., they are degenerate and hence, are represented by an arbitrary variable (in this paper it is designated by q) or '0' as mentioned in certain portions of the literature.

The TE and TM modes of a spherical dielectric body are further categorized into external and internal modes. While the internal modes mostly determine the field configurations within the scattering dielectric sphere, the external modes

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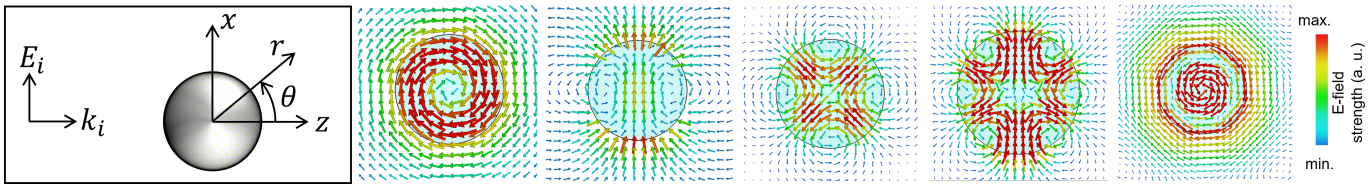


Fig. 1. E-field configuration (eigenmode simulations in CST Microwave Studio) of natural electromagnetic modes of a homogeneous dielectric sphere with plane wave incidence. From left to right, they are magnetic dipole, electric dipole, electric quadrupole, electric octopole, and magnetic quadrupole, respectively. The color scale changes from blue at minimum to red at maximum amplitude.

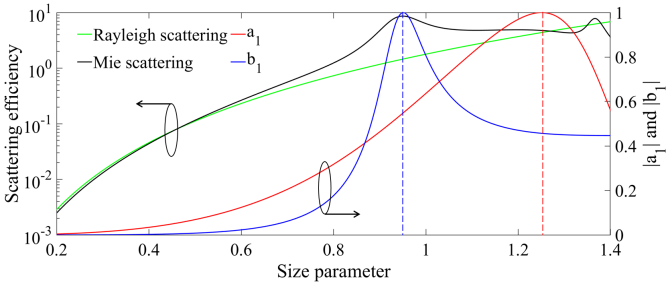


Fig. 2. Theoretical scattering efficiency and Mie coefficients of a dielectric sphere of $\varepsilon_r = 10$ immersed in infinite free space.

are concerned with the scattered field outside the dielectric sphere. Mathematically, the electric and magnetic field strength of these modes are calculated using the Mie coefficients a_n, b_n, c_n, and d_n, which represent the external TM mode, external TE mode, internal TE mode, and internal TM mode, respectively. The dominant electromagnetic mode of the dielectric sphere of relative permittivity ε_r suspended in dielectric medium of relative permittivity ε_m is the magnetic dipole mode (TE_{1q1}) when $|\epsilon_r/\epsilon_m| > 1$ and the electric dipole mode (TM_{1q1}) for $|\varepsilon_r/\varepsilon_m| < 1$ [16], [17]. While the former is a very realistic and widely used configuration of any generic dielectric resonator in free space, the latter can be depicted as an air bubble in a liquid medium. It has been proposed to characterize the electrical properties of liquids at mmWave frequency [18].

For a small size parameter $x \ll 1$ and additionally when $mx < 1$, the electric dipole mode (TM_{1q1}) is dominant, and hence, both Rayleigh and Mie scattering are almost identical. This is because the comparative strength of magnetic dipole mode (TE_{1q1}) is almost negligible in this regime. However, with increasing size parameter, the magnitude of the magnetic dipole mode increases and causes the Mie scattering and Rayleigh scattering characteristics to split away from each other. The maximum mismatch between them appears when the magnitude of the magnetic dipole coefficient attains its maximum value $|b_1| = 1$ at $x = 0.95$ (see the blue discontinuous vertical line in Fig. 2). Mathematically, this local maximum is defined as the resonance of the magnetic dipole mode, while physically, it implies maximum scattered power from the magnetic dipole mode. Similarly, $|a_1| = 1$ at $x = 1.25$, defines the electric dipole resonance (see the red discontinuous vertical line in Fig. 2).

The resonance of the natural Mie modes can be experimentally validated through metal waveguide transmission-based measurements. In the presence of a dielectric sphere placed almost symmetrically in the rectangular cross-section of a straight metal waveguide, the perturbation caused by the resonances of the Mie modes causes either transmission-zero or a reflection-zero in an end-to-end measurement [19], [20]. Such a measurement methodology is limited to the first few modes owing to the complexity in instrumentation and assembly of a dielectric sphere (diameter in the sub-millimeter regime) inside the rectangular metal waveguides at very high frequencies [21]. Alternatively, through near-field probing of the resonance scattering of E-fields, the fundamental dipole modes have been identified at terahertz frequency for a dielectric sphere of diameter in the micrometer range [22].

For this work, Alumina dielectric spheres were obtained from Goodfellow Cambridge Ltd. Under the general assumption of the dielectric constant of Alumina being approximately 10, the resonances of the fundamental dipole modes, in accordance with Mie theory are expected for size parameter between 0.9 to 1.3 (see Fig. 1). With the diameter of the sphere being 3 mm (with a tolerance of $\pm 2.5\,\mu$m and sphericity of 0.625 μm), the size parameter range translates to a test frequency between 28 GHz to 40 GHz. Hence, a WR-28 rectangular metal waveguide, operating in the fundamental TE_{10} mode, is chosen to excite the electric and magnetic dipole modes. The dielectric sphere is positioned in the center of the H-plane of the rectangular cross-section of the metal waveguide (see Fig. 3 [top left]). However, in the E-plane, a precise centering is not practically feasible (see the dimensions in Fig. 3 [top right]). Instead, for simplicity, the sphere is made to rest on the bottom wall of the waveguide. Furthermore, this robust packaging helps to generate a deterministic 3D modeling of the test setup in the electromagnetic solver for field simulations. Necessary foam-based packaging materials ($\varepsilon_r \approx 1$) are used to affix the sphere in its desired location inside the waveguide (see photo in Fig. 3 [center right]).

For two-port S-parameter measurements, 2.92 mm coax-to-waveguide adapters are used (see in Fig. 3 [center left]). An open-short-load-thru calibration is done at the coaxial terminals. First, a full 2-port measurement is recorded with the packaging foam only for the de-embedding of the measured

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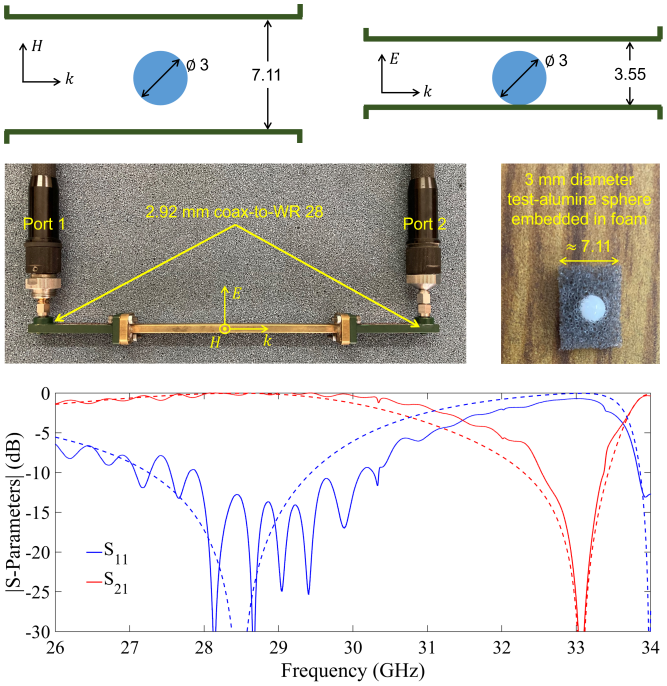


Fig. 3. Design of the experimental setup with results: location of the sphere in the [top left] H-plane and in the [top right] E-plane of the WR-28; [center left] photo of the 2-port measurement system with [center right] the sphere embedded in a foam platform; [bottom] measured response (continuous lines) along with fitted simulated data (discontinuous lines). All geometrical dimensions shown in this figure are in millimeters.

data with foam and sphere. The final results after removing the loss from the foam package and coax-to-waveguide adapters are shown in Fig. 3 [bottom]. The simulated results are fitted to the measured response by optimizing the dielectric properties of the sphere. A good agreement can be seen for both the input-reflection and through-transmission parameters. The corresponding optimized dielectric parameters are $\varepsilon_r = 10.5$ and $\tan\delta \approx 10^{-3}$. Note that these numerical values obtained through full-wave simulations are subject to a finite uncertainty, one primary reason being the sphericity and tolerance of the sphere, which is critical to include in the 3D-CAD model of the electromagnetic solver.

The magnetic dipole mode, as identified by its circumferential (non-radial) E-field distribution and simultaneously an almost linear H-field configuration inside the sphere, is excited by the incident TE_{10} mode of the metal waveguide (see in Fig. 4a). In the E-plane (Fig. 4a [center]), the E-field configuration of the TE_{10} transmission of the metal waveguide is aligned with the TE_{1q1} resonance of the dielectric sphere in accordance to the boundary conditions at the air-dielectric interface. Also, in the H-plane (Fig. 4a [bottom]), the magnetic-symmetry plane is conserved for both the TE_{10} transmission of the metal waveguide and the TE_{1q1} resonance of the dielectric sphere. Hence, a perfect transmission across the dielectric sphere with a reflection zero is detected around the resonance of the magnetic dipole mode, which in this case is around 28.5 GHz.

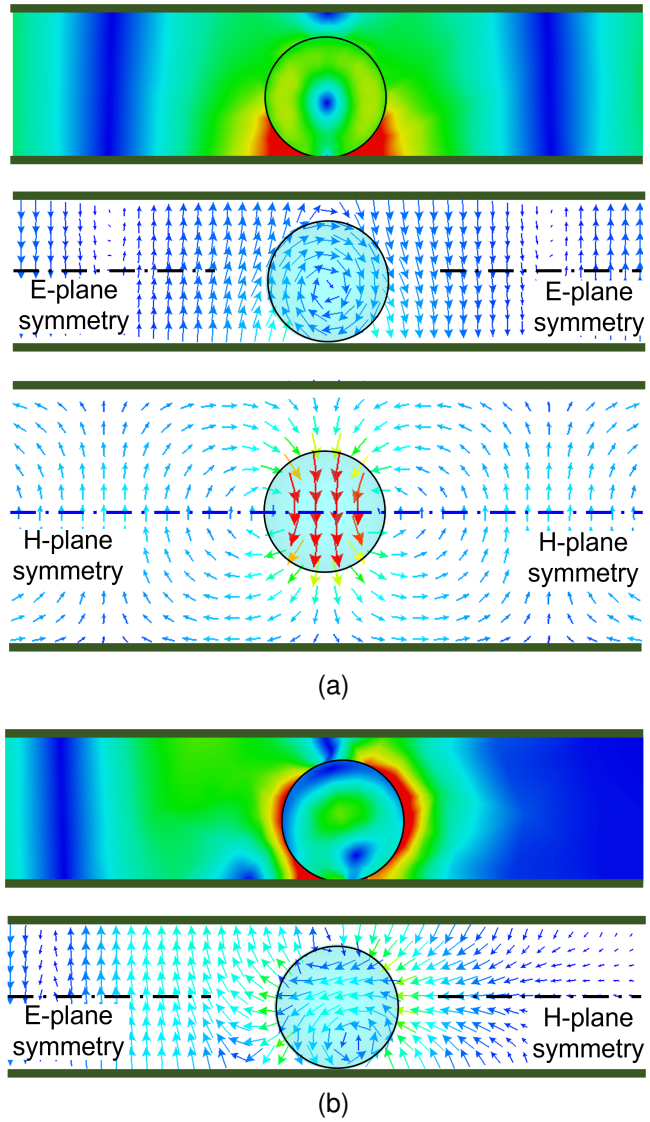


Fig. 4. Simulated field distribution (CST Microwave Studio) of the experimental setup as demonstrated in Fig. 3. Note the color scale changes from blue at minimum to red for the maximum magnitude of the fields. (a) [top] E-field intensity, [center] circumferential E-field configuration, and [bottom] linear-polarized H-field configuration at 28.5 GHz revealing the resonant state of the magnetic dipole mode. (b) [top] E-field intensity and [bottom] E-field configuration at 33.1 GHz depicting a perturbation state of the resonating electric dipole mode.

Alternatively, a transmission-zero is measured around 33.1 GHz for the electric dipole mode. The incident TE_{10} mode in the rectangular metal waveguide excites the electric dipole moment in the dielectric sphere, which in turn induces a magnetic-symmetry plane (see in Fig. 4b). Thus, the propagation of the TE_{10}, which must follow an electric-symmetry plane, cease-to-exist. Instead, the electric dipole resonance mode converts the incident TE_{10} mode to the TM_{11} mode of the rectangular metal waveguide, which aligns with the magnetic-symmetry plane. However, the cut-off frequency of the TM_{11} mode in WR-28 is 47.19 GHz, thereby being in the evanescent state in the current operating frequency regime.

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This explains the transmission zero and reflection coefficient reaching almost unity.

III. ASSEMBLY AND FEEDING OF SPHERICAL DIELECTRIC RESONATOR ON A PLANAR PLATFORM

The total energy (W) associated with any given mode consists of three parts: the energy stored inside the sphere (W_1), reactive energy stored outside the sphere (W_2), and scattered energy in far-field region (W_3). By virtue of a magnetic symmetry plane induced within a parallel metal plane configuration, the radiation (TEM mode) from the resonating sphere is curtailed (see the resultant field configurations in Fig. 5a). Note that both the electric and magnetic dipole modes present an H-plane symmetry (marked by a blue line) midway between the two parallel conductor layers. Henceforth, the Q-factor of the spherical dielectric resonator is now exclusively dependent on dielectric dissipation loss in the sphere (when $\tan\delta \neq 0$). Using Mie internal coefficients, c_1 and d_1, mathematically it has been established that [20]

$$W_1^{H_{nqr}} = \frac{W_1^{E_{nqr}}}{m'^2 - m''^2} \tag{3}$$

where m' and m'' are real and imaginary parts, respectively of the dielectric sphere's refractive index (m) immersed in infinite free space. Hence, the dielectric filling factor (W_1/W) of the TM_{1q1} is higher than TE_{1q1}. So, compared to the TE_{1q1} mode, the TM_{1q1} exhibits higher dielectric loss in the sphere and, thus, a lower Q-factor. The higher dielectric loss for the TM_{1q1} mode is also revealed in the absorption efficiency as shown in Fig. 5b. Note that with an increase in loss tangent of the dielectric sphere, the difference between absorption efficiency of the TM_{1q1} and TE_{1q1} decreases according to Eq. 3.

For experimental verification, a K_a-band scaled prototype (see Fig. 5c) is designed on Rogers 3003 substrate ($\varepsilon_r = 3$, $\tan\delta = 0.001$ at 10 GHz) of thickness 0.14 mm. As discussed before (section I), the circular/square cavity is drilled into the thin low-permittivity substrate layer that allows for self-alignment of the sphere with extreme placement accuracy on the chip at mmWave frequency. The circular crate for placement of the alumina sphere has a diameter of 0.4 mm. The gap between the crate from the 50 Ω microstrip feedline (width 0.125 mm) is 1.5 mm. The alumina sphere of diameter 3 mm is used in K_a-band prototype (as discussed in the previous section). The fabricated design and measured results are shown in Fig. 5c and Fig. 5d, respectively.

The fringing electric field from the microstrip feedline excites the TE_{1q1} and TM_{1q1} modes at around 33.02 GHz and 34.47 GHz, respectively. The extracted unloaded Q-factor from measurements (see in Fig. 5d) are around 2352 and 1363 for the TE_{1q1} and TM_{1q1} modes, respectively. Thus, the TE_{1q1} features a higher internal Q-factor than the TM_{1q1} mode, which was expected according to the theoretical background discussed before. Hence, for design on on-chip circuits like resonators, narrowband filters [23], and sensors [24], the

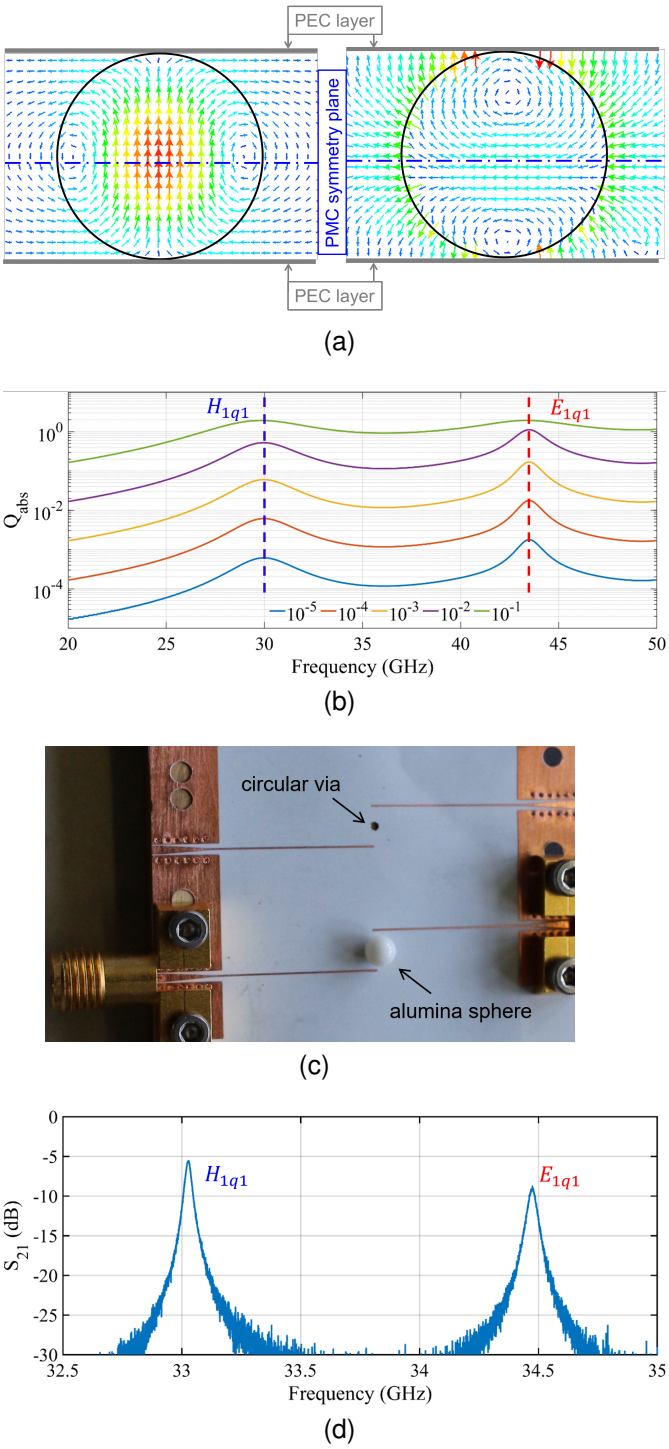


Fig. 5. Design, simulations (CST Microwave Studio), and measurements of a frequency-scaled prototype for an on-chip resonator. (a) Simulated [left] H-field configuration of the magnetic dipole mode and [right] E-field configuration of the electric dipole mode in the dielectric sphere sandwiched between two parallel metal plates. (b) Theoretical absorption efficiency calculated from Mie coefficient for a dielectric sphere of diameter 3 mm and dielectric constant 10.3, suspended in infinite free space with varying loss tangent. (c) Photo of the assembled circuit design for measurements in K_a-band. Note that the top metal plate is not shown in the photo of the hardware. (d) Measured results revealed the resonances of the two fundamental dipole modes.

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TE_{1q1} is desirable. Alternatively, for design of antennas requiring low-Q-factor [25], and wideband circuits (for e.g., electromagnetic transition design [26]), the TM_{1q1} will be instrumental.

The product of Q-factor and resonance frequency achieved by the TE_{1q1} mode is in the order of 10^5 GHz. Similar level of performance ($Q \times f_0$) has also been achieved by the use of cylindrical dielectric resonators in [27], [28]. However, the minimal on-chip footprint presented by a spherical geometry is its figure of merit compared to a traditional cylindrical-shaped dielectric resonator with the same diameter. For example, the on-chip area consumption (in terms of λ_0^2, where λ_0 is free space wavelength at operational resonance frequency) for the designs presented in [27], [28] are approximately ten times more than the circuit design shown in the current work.

IV. Conclusion

The information-rich electromagnetic scattering spectrum of dielectric spheres has been explored theoretically about a century ago. However, such formulations have not been extensively exploited for applications in electrical engineering. In this context, the current work analyzes the fundamental dipole modes of a spherical dielectric resonator and explores their potential features for high-frequency circuit designs. The adaptation of the fundamental dipole modes of Alumina ceramic-based spherical dielectric resonator for the development of high-Q on-chip circuits is demonstrated analytically and experimentally between 30 GHz to 35 GHz.

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